

Olist E-Commerce Performance Report

Advanced SQL Analytics & Business Intelligence Insights

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Tooling: Python, MySQL, Pandas, SQLAlchemy

Dataset: Brazilian E-Commerce Public Dataset by Olist

1. Executive Summary

This report details the technical analysis of 100k+ anonymized orders from Olist, a major Brazilian marketplace. The focus of this analysis is twofold: evaluating the **financial efficiency** of logistics through freight ratios and tracking **macro-economic growth** via Month-over-Month (MoM) revenue trends.

2. Technical Workflow

The project utilized a robust data pipeline:

- **Data Engineering:** Python (Pandas) was used to merge multiple relational tables, handle null values in product dimensions, and translate category names into English.
- **Database Integration:** The cleaned dataframe was pushed to a MySQL local instance using SQLAlchemy for advanced querying.
- **Analysis:** SQL CTEs and Window Functions were employed to derive complex business metrics.

3. Key Insight: Freight-to-Price Efficiency

To assess profitability, we analyzed the ratio of shipping costs (Freight) to the item price. High ratios indicate categories where logistics costs significantly eat into margins.

Finding: Categories like bulky furniture and heavy appliances often show a ratio exceeding 25%, suggesting a need for regional warehousing optimization.

```
-- Freight-to-Price Ratio Query
SELECT
    product_category_name_english,
    ROUND(SUM(price), 2) AS total_sales,
    ROUND(SUM(freight_value), 2) AS total_freight,
    ROUND((SUM(freight_value) / SUM(price)) * 100, 2) AS freight_ratio_pct
FROM olist_data
GROUP BY product_category_name_english
ORDER BY freight_ratio_pct DESC
LIMIT 5;
```

4. Growth Trends: Month-over-Month Revenue

The marketplace growth was tracked using time-series analysis to identify seasonal peaks and performance volatility.

```
-- MoM Growth Query
WITH MonthlySales AS (
    SELECT
        DATE_FORMAT(shipping_limit_date, '%Y-%m') AS order_month,
        ROUND(SUM(price), 2) AS monthly_revenue
    FROM olist_data
    GROUP BY order_month
)
SELECT
    order_month,
    monthly_revenue,
    LAG(monthly_revenue) OVER (ORDER BY order_month) AS last_month_revenue,
    ROUND(((monthly_revenue - LAG(monthly_revenue) OVER (ORDER BY order_month)) /
        LAG(monthly_revenue) OVER (ORDER BY order_month)) * 100, 2) AS growth_pct
FROM MonthlySales;
```

5. Strategic Recommendations

1. **Logistics Optimization:** Focus on reducing the freight-to-price ratio for high-volume but heavy categories through local fulfillment centers.
2. **Seller Loyalty:** Implement incentives for sellers contributing to high MoM growth periods to ensure stock availability during peaks.

3. **Retention Focus:** Utilize the SQL-derived customer segments for targeted email marketing campaigns.